

Software Design Patterns Documentation (with Code Snippets)

Project: Skill Verse — LMS with Intelligent Task Planner

Course: CSE 327 Software Engineering

University: North South University

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Executive Summary

This document presents a comprehensive analysis of the **10 Software Design Patterns** implemented throughout the **Skill Verse LMS** codebase. The patterns are categorized according to classic Gang of Four (GoF) and Modern Architectural conventions into:

- **Creational Patterns** (Factory Method, Singleton)
- **Structural Patterns** (Facade, Decorator/Wrapper, Provider/Context)
- **Behavioral Patterns** (Strategy, State, Observer, Chain of Responsibility)
- **Architectural Patterns** (Repository)

Summary Matrix

#	Pattern Name	Category	Primary Files	Key Responsibility
1	Factory Method	Creational	[<code>utils/supabase/client.ts</code>] (file:///c:/Users/Nafiz%20Imtiaz/Desktop/lms_with_intelligent_task_planner-master/utils/supabase/client.ts) [<code>utils/supabase/server.ts</code>] (file:///c:/Users/Nafiz%20Imtiaz/Desktop/lms_with_intelligent_task_planner-master/utils/supabase/server.ts)	Encapsulates client vs server Supabase object creation.
2	Singleton	Creational	[<code>utils/supabase/client.ts</code>] (file:///c:/Users/Nafiz%20Imtiaz/Desktop/lms_with_intelligent_task_planner-master/utils/supabase/client.ts) [<code>lib/google-calendar.ts</code>] (file:///c:/Users/Nafiz%20Imtiaz/Desktop/lms_with_intelligent_task_planner-master/lib/google-calendar.ts)	Maintains a single client bundle connection instance across component renders.
3	Observer	Behavioral	[<code>app/learner/canvas/assignments/page.tsx</code>] (file:///c:/Users/Nafiz%20Imtiaz/Desktop/lms_with_intelligent_task_planner-master/app/learner/canvas/assignments/page.tsx#L62-L87) [<code>hooks/use-speech-to-text.ts</code>] (file:///c:/Users/Nafiz%20Imtiaz/Desktop/lms_with_intelligent_task_planner-master/hooks/use-speech-to-text.ts)	Subscribes UI components to database changes and speech event streams.
4	Strategy	Behavioral	[<code>components/ai-teaching-assistant.tsx</code>] (file:///c:/Users/Nafiz%20Imtiaz/Desktop/lms_with_intelligent_task_planner-master/components/ai-teaching-assistant.tsx) [<code>utils/file-parser.ts</code>] (file:///c:/Users/Nafiz%20Imtiaz/Desktop/lms_with_intelligent_task_planner-master/utils/file-parser.ts)	Switches algorithms dynamically based on mode or file type at runtime.
5	State	Behavioral	[<code>app/api/ai/interview/route.ts</code>] (file:///c:/Users/Nafiz%20Imtiaz/Desktop/lms_with_intelligent_task_planner-master/app/api/ai/interview/route.ts#L138-L161)	Transitions AI behavior through interview stages (<code>OPENING</code> → <code>ONGOING</code> → <code>LATE</code> → <code>CLOSING</code>).
6	Facade	Structural	[<code>components/ai-teaching-assistant.tsx</code>] (file:///c:/Users/Nafiz%20Imtiaz/Desktop/lms_with_intelligent_task_planner-master/components/ai-teaching-assistant.tsx) [<code>lib/canvas-api.ts</code>] (file:///c:/Users/Nafiz%20Imtiaz/Desktop/lms_with_intelligent_task_planner-master/lib/canvas-api.ts)	Provides a unified interface hiding multi-endpoint REST and AI API complexity.

7	Provider / Context	Structural	[components/auth-context.tsx] (file:///c:/Users/Nafiz%20Imtiaz/Desktop/lms_with_intelligent_task_planner-master/components/auth-context.tsx)	Injects authentication state and methods down the component tree without prop drilling.
8	Decorator / Wrapper	Structural	[components/learner-layout.tsx] (file:///c:/Users/Nafiz%20Imtiaz/Desktop/lms_with_intelligent_task_planner-master/components/learner-layout.tsx) [components/educator-layout.tsx] (file:///c:/Users/Nafiz%20Imtiaz/Desktop/lms_with_intelligent_task_planner-master/components/educator-layout.tsx)	Wraps raw view components with sidebar navigation, header, and auth enforcement.
9	Chain of Responsibility	Behavioral	[middleware.ts] (file:///c:/Users/Nafiz%20Imtiaz/Desktop/lms_with_intelligent_task_planner-master/middleware.ts)	Sequentially evaluates authentication, user roles, and route permissions per request.
10	Repository	Architectural	[app/api/tasks/route.ts] (file:///c:/Users/Nafiz%20Imtiaz/Desktop/lms_with_intelligent_task_planner-master/app/api/tasks/route.ts) [app/api/canvas/courses/route.ts] (file:///c:/Users/Nafiz%20Imtiaz/Desktop/lms_with_intelligent_task_planner-master/app/api/canvas/courses/route.ts)	Encapsulates database queries behind standardized REST API handlers.

1. Factory Method Pattern (Creational)

Intent

Defines an interface or function for creating objects, allowing subclasses or module functions to decide which class/instance to instantiate based on execution context (e.g., Browser vs Server).

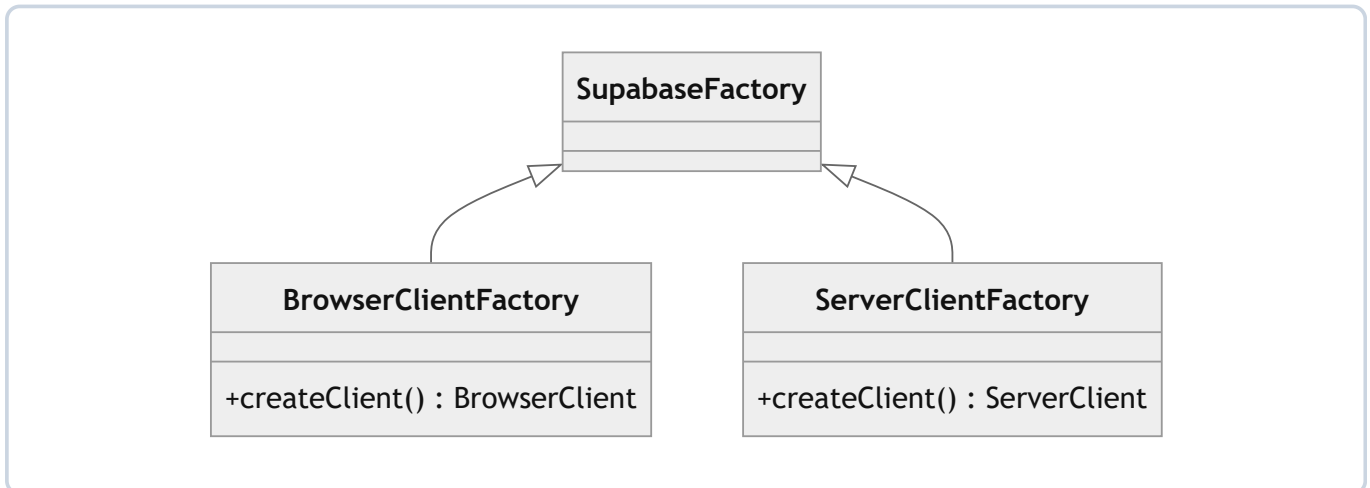
Implementation Details

In Next.js 15 App Router, Supabase client initialization differs depending on execution context:

- **Browser Context:** Needs `createBrowserClient` with client environment variables.
- **Server Context:** Needs `createServerClient` with asynchronous Next.js cookie handling (`cookies()`).

Both implement a unified `createClient()` interface factory function.

Class Diagram



Code Snippets

Browser Factory — [[utils/supabase/client.ts](#)]

(file:///c:/Users/Nafiz%20Imtiaz/Desktop/lms_with_intelligent_task_planner-master/utils/supabase/client.ts#L1-L10)

```
import { createBrowserClient } from '@supabase/ssr'

export function createClient() {
  return createBrowserClient(
    process.env.NEXT_PUBLIC_SUPABASE_URL!,
    process.env.NEXT_PUBLIC_SUPABASE_PUBLISHABLE_KEY!
  )
}
```

Server Factory — [[utils/supabase/server.ts](#)]

(file:///c:/Users/Nafiz%20Imtiaz/Desktop/lms_with_intelligent_task_planner-master/utils/supabase/server.ts#L1-L29)

```
import { createServerClient } from '@supabase/ssr'
import { cookies } from 'next/headers'

export async function createClient() {
  const cookieStore = await cookies()

  return createServerClient(
    process.env.NEXT_PUBLIC_SUPABASE_URL!,
    process.env.NEXT_PUBLIC_SUPABASE_PUBLISHABLE_KEY!,
    {
      cookies: {
        getAll() {
          return cookieStore.getAll()
        },
        setAll(cookiesToSet) {
          try {
            cookiesToSet.forEach(({ name, value, options }) =>
              cookieStore.set(name, value, options)
            )
          } catch {
            // Ignored if called from Server Component
          }
        },
      },
    },
  )
}
```

2. Singleton Pattern (Creational)

Intent

Ensures a class or client instance has only one global instance throughout the user session and provides a global access point to it.

Implementation Details

- `createBrowserClient` internally memoizes and reuses a single connection instance within the browser window runtime to avoid redundant connection overhead.
- `initGoogleCalendarService` initializes a single `GoogleCalendarService` instance per user session, preventing multiple OAuth authentication handshakes.

 Code Snippet — [[lib/google-calendar.ts](#)]

(file:///c:/Users/Nafiz%20Imtiaz/Desktop/lms_with_intelligent_task_planner-master/lib/google-calendar.ts#L320-L340)

```

export async function initGoogleCalendarService(): Promise {
  const supabase = await createClient()
  const { data: { user } } = await supabase.auth.getUser()

  if (!user) return null

  const { data: connection } = await supabase
    .from('calendar_sync_settings')
    .select('access_token, refresh_token')
    .eq('user_id', user.id)
    .single()

  if (!connection?.access_token || !connection?.refresh_token) {
    return null
  }

  return new GoogleCalendarService(connection.access_token, connection.refresh_token)
}

```

3. Observer Pattern (Behavioral)

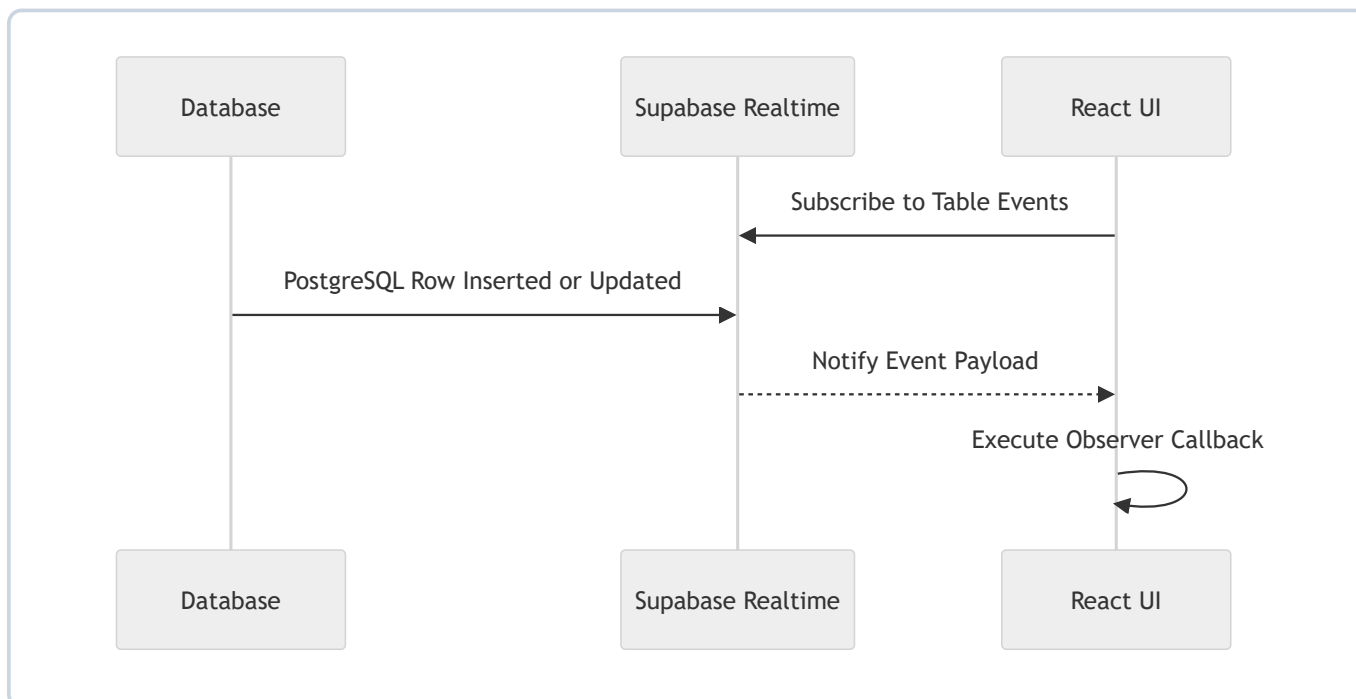
 Intent

Defines a one-to-many dependency between objects so that when one object changes state (e.g., database update, speech recognition event, deadline timer), all its dependents are notified automatically.

 Implementation Details

1. **Supabase Realtime WebSockets:** The React client subscribes to `postgres_changes` on the `canvas_assignments` table. When an assignment row is updated or inserted in PostgreSQL, Supabase notifies the client subscriber, which automatically calls `fetchData()`. 2. **Web Speech API:** `useSpeechToText` attaches callbacks (`onresult`, `onend`, `onerror`) to the browser's `SpeechRecognition` instance.

Sequence Diagram



Code Snippet — [[app/learner/canvas/assignments/page.tsx](#)]

(file:///c:/Users/Nafiz%20Imtiaz/Desktop/lms_with_intelligent_task_planner-master/app/learner/canvas/assignments/page.tsx#L62-L87)

```

// Real-time subscription for canvas_assignments updates
useEffect(() => {
  const supabase = createClient()

  // Subscribe to changes in canvas_assignments table
  const channel = supabase
    .channel('canvas_assignments_changes')
    .on(
      'postgres_changes',
      {
        event: '*', // Listen to INSERT, UPDATE, DELETE
        schema: 'public',
        table: 'canvas_assignments'
      },
      (payload) => {
        console.log('Canvas assignment changed:', payload)
        // Refresh data when any assignment is updated
        fetchData()
      }
    )
    .subscribe()

  // Cleanup subscription on unmount
  return () => {
    supabase.removeChannel(channel)
  }
}, [selectedCourse, statusFilter, submittedFilter])

```

4. Strategy Pattern (Behavioral)

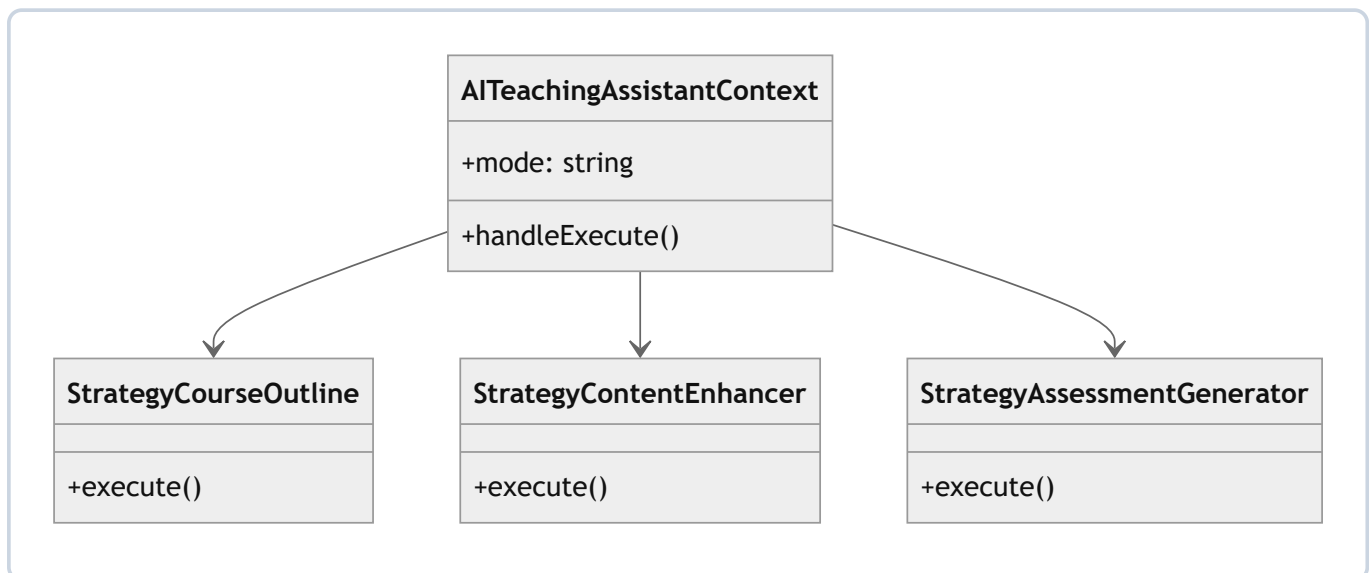
Intent

Enables selecting an algorithm or execution strategy at runtime without altering the context that uses it.

Implementation Details

- **AI Teaching Assistant** (`AITeachingAssistant`): Accepts a `mode` prop (`"course-outline"` | `"content-enhancer"` | `"assessment-generator"` | `"student-insights"`). Based on `mode` , it selects different API endpoints, parameters, system prompts, and UI render strategies.
- **File Parser** (`extractTextFromFile`): Evaluates file type (`image/*` vs text) and dispatches to the appropriate parsing strategy (`extractTextFromImage` via Tesseract OCR).

Strategy Architecture



 Code Snippet — [components/ai-teaching-assistant.tsx]

(file:///c:/Users/Nafiz%20Imtiaz/Desktop/lms_with_intelligent_task_planner-master/components/ai-teaching-assistant.tsx#L23-L80)

```

interface AITeachingAssistantProps {
  mode: "course-outline" | "content-enhancer" | "assessment-generator" | "student-insights"
  onClose?: () => void
  onApply?: (data: any) => void
  initialData?: any
}

export function AITeachingAssistant({ mode, initialData }: AITeachingAssistantProps) {
  // Strategy execution based on mode parameter
  const handleGenerate = () => {
    switch (mode) {
      case "course-outline":
        return handleGenerateCourseOutline()
      case "content-enhancer":
        return handleEnhanceContent()
      case "assessment-generator":
        return handleGenerateAssessment()
      case "student-insights":
        return handleFetchInsights()
    }
  }
}

```

5. State Pattern (Behavioral)

Intent

Allows an object to alter its behavior when its internal state changes, making it appear as if the object changed its class.

Implementation Details

In the **AI Interview Engine** ([app/api/ai/interview/route.ts]

(file:///c:/Users/Nafiz%20Imtiaz/Desktop/lms_with_intelligent_task_planner-master/app/api/ai/interview/route.ts#L138-L161)), the interview session transitions through explicit stages based on the number of completed questions (`questionCount`): 1. **OPENING** — Welcomes the candidate and asks introductory background questions. 2. **ONGOING** — Asks deep technical and domain-specific questions based on the job description. 3. **LATE** — Evaluates problem-solving and scenario-based responses. 4. **CLOSING** — Wraps up professionally and generates comprehensive performance feedback.

 **Code Snippet** — [`app/api/ai/interview/route.ts`]

(file:///c:/Users/Nafiz%20Imtiaz/Desktop/lms_with_intelligent_task_planner-master/app/api/ai/interview/route.ts#L138-L161)

```
const questionCount = conversationHistory.filter((msg: any) => msg.role === "assistant").length

let stageInstruction = ""
if (questionCount >= 7) {
  stageInstruction = "\n\nINTERVIEW STAGE: CLOSING - This should be your final question or wrap up the interview professionally. Thank the candidate and indicate the interview is complete."
} else if (questionCount >= 5) {
  stageInstruction = "\n\nINTERVIEW STAGE: LATE - You're near the end. Ask 1-2 more meaningful questions before wrapping up."
} else {
  stageInstruction = "\n\nINTERVIEW STAGE: ONGOING - Continue with relevant questions based on the job requirements and previous answers."
}

// System prompt dynamically incorporates the current state instruction
const systemPrompt = You are a professional technical interviewer... ${stageInstruction}
```

6. Facade Pattern (Structural)

Intent

Provides a unified, high-level interface to a set of interfaces in a subsystem, making the subsystem easier to use.

Implementation Details

- **CanvasAPIService** : Encapsulates multiple complex low-level Canvas REST endpoints (`/api/v1/users/self` , `/api/v1/courses` , `/api/v1/announcements` , `/api/v1/users/self/enrollments?include[]=grades`) into single high-level clean method calls like `getCourses()` , `getAllAnnouncements()` , and `getEnrollments()` .

 Code Snippet — [lib/canvas-api.ts]

(file:///c:/Users/Nafiz%20Imtiaz/Desktop/lms_with_intelligent_task_planner-master/lib/canvas-api.ts#L72-L105)

```

export class CanvasAPIService {
  private canvasUrl: string
  private accessToken: string
  private baseUrl: string

  constructor(config: CanvasConfig) {
    this.canvasUrl = config.canvasUrl.replace(/\/$/, '')
    this.accessToken = config.accessToken
    this.baseUrl = `${this.canvasUrl}/api/v1`
  }

  // Facade method hiding HTTP headers, bearer tokens, and endpoint formatting
  private async request(endpoint: string, options: RequestInit = {}): Promise {
    const url = `${this.baseUrl}${endpoint}`
    const response = await fetch(url, {
      ...options,
      headers: {
        'Authorization': `Bearer ${this.accessToken}`,
        'Content-Type': 'application/json',
        ...options.headers,
      },
    })
    return response.json()
  }

  // Simplified high-level facade interface used by application
  async getCourses(): Promise {
    return this.request('/courses?include[]=term&per_page=100')
  }
}

```

7. Provider / Context Pattern (Structural / Dependency Injection)

Intent

Provides a way to share values (such as authentication state and user details) between components without having to explicitly pass a prop through every level of the component tree.

Implementation Details

`AuthProvider` wraps the React root tree and injects `user`, `isLoading`, `signIn`, `signUp`, and `signOut` into any child component using the custom `useAuth()` hook.

 **Code Snippet** — [components/auth-context.tsx]

(file:///c:/Users/Nafiz%20Imtiaz/Desktop/lms_with_intelligent_task_planner-master/components/auth-context.tsx#L24-L112)

```
const AuthContext = createContext(undefined)

export function AuthProvider({ children }: { children: React.ReactNode }) {
  const [user, setUser] = useState(null)
  const [isLoading, setIsLoading] = useState(true)

  return (

    {children}

  )
}

export function useAuth() {
  const context = useContext(AuthContext)
  if (context === undefined) {
    throw new Error("useAuth must be used within an AuthProvider")
  }
  return context
}
```

8. Decorator / Wrapper Pattern (Structural)

Intent

Attaches additional responsibilities or layout capabilities to an object/component dynamically without altering its core structure.

Implementation Details

`LearnerLayout` and `EducatorLayout` act as structural decorators. They accept raw page components (`children`) and dynamically wrap them with sidebar navigation, mobile drawer toggles, theme styling, and header user controls.

 Code Snippet — [components/learner-layout.tsx]

(file:///c:/Users/Nafiz%20Imtiaz/Desktop/lms_with_intelligent_task_planner-master/components/learner-layout.tsx#L24-L50)

```

interface LearnerLayoutProps {
  children: React.ReactNode
}

export function LearnerLayout({ children }: LearnerLayoutProps) {
  return (

    {/* Sidebar Decorator */}

    {/* Main Decorated Content */}

    {children}

  )
}

```

9. Chain of Responsibility Pattern (Behavioral)

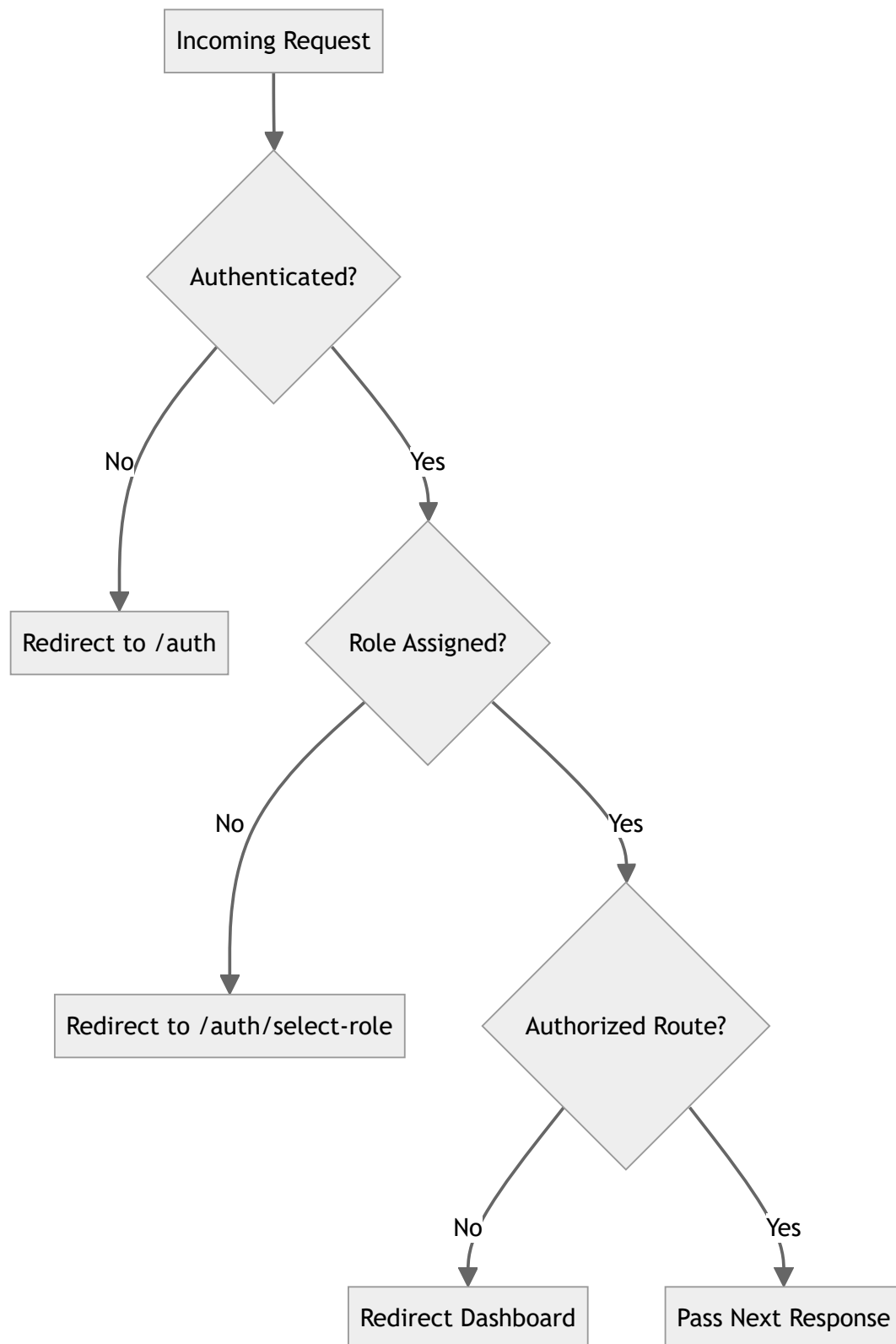
Intent

Avoids coupling the sender of a request to its receiver by giving more than one object a chance to handle the request. Passes the request along a chain until an object handles or redirects it.

Implementation Details

In Next.js middleware ([middleware.ts](file:///c:/Users/Nafiz%20Imtiaz/Desktop/lms_with_intelligent_task_planner-master/middleware.ts#L4-L109)), every incoming HTTP request passes through a sequential validation chain: 1. **Authentication Guard**: Is user logged in? If not and route is protected → redirect to `/auth`. 2. **Role Verification**: Does user profile have a role? If missing → redirect to `/auth/select-role`. 3. **Role Authorization**: Is a learner accessing `/educator/*` or educator accessing `/learner/*`? If mismatched → redirect to correct role dashboard. 4. **Auth Page Redirection**: Is an authenticated user accessing `/auth`? → redirect to dashboard.

Chain Flow Diagram



 Code Snippet — [`middleware.ts`]

(file:///c:/Users/Nafiz%20Imtiaz/Desktop/lms_with_intelligent_task_planner-master/middleware.ts#L54-L89)

```

// 1. Authentication Check
if (!user && isProtectedRoute) {
  const redirectUrl = new URL('/auth', request.url)
  redirectUrl.searchParams.set('redirectTo', path)
  return NextResponse.redirect(redirectUrl)
}

// 2. Role Check & Authorization Check
if (user && (isEducatorRoute || isLearnerRoute)) {
  const { data: profile } = await supabase
    .from('profiles')
    .select('role')
    .eq('id', user.id)
    .single()

  if (!profile?.role) {
    return NextResponse.redirect(new URL('/auth/select-role', request.url))
  }

  if (isEducatorRoute && profile.role !== 'educator') {
    return NextResponse.redirect(new URL('/learner/dashboard', request.url))
  }

  if (isLearnerRoute && profile.role !== 'learner') {
    return NextResponse.redirect(new URL('/educator/dashboard', request.url))
  }
}

return supabaseResponse

```

10. Repository Pattern (Architectural)

Intent

Mediates between the domain and data mapping layers using a collection-like interface for accessing domain objects, isolating database access logic from presentation components.

Implementation Details

Next.js Server API routes (`app/api/*`) encapsulate Supabase PostgreSQL data queries. Client components never perform raw database queries directly for task creation or updates; instead, they interact through standardized API endpoints (`/api/tasks` , `/api/canvas/courses`).

Code Snippet — [[app/api/tasks/route.ts](#)]

(file:///c:/Users/Nafiz%20Imtiaz/Desktop/lms_with_intelligent_task_planner-master/app/api/tasks/route.ts#L19-L37)

```
export async function GET() {
  const supabase = await createClient()
  const { data: { user }, error: authError } = await supabase.auth.getUser()

  if (authError || !user) {
    return NextResponse.json({ error: 'Unauthorized' }, { status: 401 })
  }

  // Repository data retrieval abstraction
  const { data: tasks, error } = await supabase
    .from('learner_tasks')
    .select('*', course:courses(id, title))
    .eq('learner_id', user.id)
    .order('created_at', { ascending: false })

  return NextResponse.json({ tasks })
}
```

Presentation Conclusion

By employing these **10 Design Patterns**, the **Skill Verse LMS** architecture achieves:

- **High Cohesion & Low Coupling** (Repository, Facade, Factory Method)
- **Extensibility & Clean Code** (Strategy, State, Observer)
- **Robust Security & Type Safety** (Chain of Responsibility Middleware, Provider Context)